

- The function $f(x; W)$ has at least two fixed points which can be found by the fixed-point iteration, and we denoted them as p_1, p_2 .
- For $i \in \{1, 2\}$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in [p_{i,1} - \epsilon_i, p_{i,1} + \epsilon_i] \times \{1\} \times \{1\}$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)}; W)$ converges to the fixed point p_i .
- For $i \in \{1, 2\}$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 1)$ such that for any $t \geq 2$,

$$\|x^{(t)} - p_i\|_\infty \leq K_i^t \cdot c_i \epsilon_i.$$

Proof. The lemma holds by combining Lemma B.1, Lemma B.2, and Lemma B.3. $\square$

Then we can further extend the Lemma B.4 to the version without dummy variables by carefully designing the weight matrix.

Lemma B.5 (A specific setting without dummy variables). *Let $W := [w_1, w_2, w_3] \in \mathbb{R}^{3 \times 3}$ be a weight matrix. Let $b := [b_1, b_2, b_3] \in \mathbb{R}^3$ be a bias matrix. Let $C = \sqrt{\frac{15+\sqrt{65}}{8}}$ be a constant. Let a polynomial activation function $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as*

$$g(z) := -\frac{2}{5}z^4 + \frac{8}{5}Cz^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)z^2 + \left(\frac{8}{5}C^3 - 3C\right)z + 1.$$

We define one layer of a neural network, denoted as f , as follows:

$$f(x; W) := [g(\langle w_1, x \rangle + b_1), g(\langle w_2, x \rangle + b_2), g(\langle w_3, x \rangle + b_3)]^\top.$$

Then there exists a weight matrix W and a bias matrix b following statements hold:

- The function $f(x; W)$ has at least two fixed points which can be found by the fixed-point iteration, and we denoted them as p_1, p_2 .
- For $i \in \{1, 2\}$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in [p_{i,1} - \epsilon_{i,1}, p_{i,1} + \epsilon_{i,1}] \times [p_{i,2} - \epsilon_{i,2}, p_{i,2} + \epsilon_{i,2}] \times [p_{i,3} - \epsilon_{i,3}, p_{i,3} + \epsilon_{i,3}]$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)}; W)$ converges to the fixed point p_i .
- For $i \in \{1, 2\}$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 1)$ such that for any $t \geq 2$,

$$\|x^{(t)} - p_i\|_\infty \leq K_i^t \cdot c_i \epsilon_i.$$

Proof. Let $w_1 := [1, 0, 0]^\top$, $w_2 := [0, 1, 0]^\top$, $w_3 := [0, 0, 1]^\top$, $b = [C, C, C]^\top$. Then it's clear that we have

$$\begin{aligned} g(\langle w_1, x \rangle + b_1) &= f_1(x) = -\frac{2}{5}x_1^4 + \frac{3}{2}x_1^2 \\ g(\langle w_2, x \rangle + b_2) &= f_2(x) = -\frac{2}{5}x_2^4 + \frac{3}{2}x_2^2 \\ g(\langle w_3, x \rangle + b_3) &= f_3(x) = -\frac{2}{5}x_3^4 + \frac{3}{2}x_3^2 \end{aligned}$$

Then combining Lemma B.1, we complete the proof. $\square$

Then, through further analysis, we can construct a neural network with polynomial activation functions that has a robust fixed point as described in Theorem 4.2. Robust fixed points ensure that small perturbations or changes in the initial conditions do not lead to significant deviations in the convergence behavior of the network. We give a 3-d version of neural network.

Lemma B.6 (A specific setting with small perturbations). *Let $W := [w_1, w_2, w_3] \in \mathbb{R}^{3 \times 3}$ be a weight matrix. Let $b := [b_1, b_2, b_3] \in \mathbb{R}^3$ be a bias matrix. Let $C = \sqrt{\frac{15+\sqrt{65}}{8}}$ be a constant. Let a polynomial activation function $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as*

$$g(z) := -\frac{2}{5}z^4 + \frac{8}{5}Cz^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)z^2 + \left(\frac{8}{5}C^3 - 3C\right)z + 1.$$

We define one layer of a neural network, denoted as f , as follows:

$$f(x; W) := [g(\langle w_1, x \rangle + b_1), g(\langle w_2, x \rangle + b_2), g(\langle w_3, x \rangle + b_3)]^\top.$$

Then there exists a weight matrix W and a bias matrix b following statements hold:

- The function $f(x; W)$ has at least two robust fixed point which can be found by the fixed-point iteration, and we denote them as p_1, p_2 ,
- For $i \in \{1, 2\}$, there exists $\epsilon_i \in \mathbb{R}^3$, for any initial point $x^{(0)} \in [p_{i,1} - \epsilon_{i,1}, p_{i,1} + \epsilon_{i,1}] \times [p_{i,2} - \epsilon_{i,2}, p_{i,2} + \epsilon_{i,2}] \times [p_{i,3} - \epsilon_{i,3}, p_{i,3} + \epsilon_{i,3}]$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)}; W)$ converges to the robust fixed point p_i .
- For $i \in \{1, 2\}$, there exists a constant $K_i \in [0, 0.9]$ and a sufficiently large constant m such that for any $t \geq 2$,

$$\|x^{(t)} - p_i\|_\infty \leq K_i^t \cdot \|\epsilon_i\|_\infty + \frac{10}{m}$$

Proof. Let m be a sufficiently large constant. Let $w_1 := [1, 1/m^2, 1/m^2]^\top$, $w_2 := [1/m^2, 1, 1/m^2]^\top$, $w_3 := [1/m^2, 1/m^2, 1]^\top$, $b = [C, C, C]^\top$. Then it's clear that we have

$$\begin{aligned} g(\langle w_1, x \rangle + b_1) &= -\frac{2}{5}x_1^4 + \frac{3}{2}x_1^2 + \frac{1}{m^2}h_1(x) \\ g(\langle w_2, x \rangle + b_1) &= -\frac{2}{5}x_2^4 + \frac{3}{2}x_2^2 + \frac{1}{m^2}h_2(x) \\ g(\langle w_3, x \rangle + b_1) &= -\frac{2}{5}x_3^4 + \frac{3}{2}x_3^2 + \frac{1}{m^2}h_3(x) \end{aligned}$$

where for $j \in [3]$, $h_j(x)$ is a polynomial in x_1, x_2, x_3 . It is obvious that when m is sufficiently large and $\|\epsilon_i\|$ is small enough, $|\frac{1}{m^2}h_j(x)| \leq \frac{1}{m}$ for $x \in D$ and $x^{(t)} \in D$ for every $t \in \mathbb{N}$. Then we can show that for any $i \in [2], j \in [3]$,

$$\begin{aligned} |x_j^{(t)} - p_{i,j}| &\leq K_{i,j}^t |x_j^{(0)} - p_{i,j}| + \frac{10}{m} \\ &\leq K_{i,j}^t \epsilon_{i,j} + \frac{10}{m} \end{aligned}$$

where the first comes from the result of Theorem 4.2, the second step comes from the range of values for the initial point. Then use the definition of $\|\cdot\|_\infty$, we have

$$\|x^{(t)} - p_i\|_\infty \leq K_i^t \cdot \|\epsilon_i\|_\infty + \frac{10}{m}$$

Then we complete the proof. $\square$

C CASE STUDY: EXPONENTIAL ACTIVATION

In this section, we extend the discussion by constructing a neural network with exponential activation functions that do not utilize dummy variables. Similar as before, through careful design, we can ensure that the network converges to different fixed points for various initialization points.

C.1 Exponential Activation

In this section, we present a univariate exponential function that has at least two fixed points, which can be identified using the fixed point methods.

Lemma C.1. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) := \exp(x^3 - 2x^2) - 1$. Then the following statements hold:*

- *The function f has at least two fixed points which can be found by the fixed-point iteration, and we denoted them as p_1, p_2 .*
- *For $i \in \{1, 2\}$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in [p_i - \epsilon_i, p_i + \epsilon_i]$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)})$ converges to the fixed point p_i .*
- *For $i \in \{1, 2\}$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 1)$ such that for any $t \geq 2$,*

$$|x^{(t)} - p_i| \leq K_i^t \cdot c_i \epsilon_i.$$

Proof. Clearly, $p_1 = 0$ and $p_2 \approx -0.9104$ are two fixed points of f . We show that p_1 and p_2 can be found by the fixed-point iteration and the error bounds hold.

For the fixed point $p_1 = 0$, let $\epsilon_1 = 0.1, K_1 = 0.5$, and $C_1 = 2/(1 - K_1)$. For any $x \in [p_1 - \epsilon_1, p_1 + \epsilon_1] = [-0.1, 0.1]$, we have $f(x) \in [-0.021, 0] \subseteq [-0.1, 0.1]$. Hence the first condition of Lemma A.4 is satisfied. For any $x \in [p_1 - \epsilon_1, p_1 + \epsilon_1]$, we have $|f'(x)| < K_1 = 0.5 < 1$. Hence the second condition of Lemma A.4 is satisfied. Thus by Lemma A.4, if we pick the initial point $x^{(0)} \in [p_1 - \epsilon_1, p_1 + \epsilon_1]$, we can find p_1 by the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, and it holds that for any $t \geq 2$,

$$\begin{aligned} |x^{(t)} - p_1| &\leq \frac{K_1^t}{1 - K_1} |x^{(1)} - x^{(0)}| \\ &\leq \frac{K_1^t}{1 - K_1} \cdot 2\epsilon_1 \\ &= K_1^t \cdot c_1 \epsilon_1, \end{aligned}$$

where the first step follows from Lemma A.4, the second step follows from $x^{(0)}, x^{(1)} \in [p_1 - \epsilon_1, p_1 + \epsilon_1]$, and the last step follows from $c_1 = 2/(1 - K_1)$.

For the fixed point $p_2 \approx -0.910$, let $\epsilon_2 = 0.1, K_2 = 0.85$, and $C_2 = 2/(1 - K_2)$. For any $x \in [p_2 - \epsilon_2, p_2 + \epsilon_2] = [-1.010, -0.810]$, we have $f(x) \in [-0.954, -0.842] \subseteq [-1.010, -0.810]$. Hence the first condition of Lemma A.4 is satisfied. For any $x \in [p_2 - \epsilon_2, p_2 + \epsilon_2]$, we have $|f'(x)| < K_2 = 0.85 < 1$. Hence the second condition of Lemma A.4 is satisfied. Thus by Lemma A.4, if we pick the initial point $x^{(0)} \in [p_2 - \epsilon_2, p_2 + \epsilon_2]$, we can find p_2 by the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, and it holds that for any $t \geq 2$,

$$|x^{(t)} - p_2| \leq \frac{K_2^t}{1 - K_2} |x^{(1)} - x^{(0)}|$$

$$\begin{aligned}
&\leq \frac{K_2^t}{1 - K_2} \cdot 2\epsilon_2 \\
&= K_2^t \cdot c_2\epsilon_2,
\end{aligned}$$

where the first step follows from Lemma A.4, the second step follows from $x^{(0)}, x^{(1)} \in [p_2 - \epsilon_2, p_2 + \epsilon_2]$, and the last step follows from $c_2 = 2/(1 - K_2)$.

Therefore, we complete the proof. $\square$

Next, we prove two lemmas which are useful to construct a neural network with a exponential activation function.

Lemma C.2. *Let*

$$f_1(x) := \exp(\langle u, x \rangle^3 + (-2 - 3C)\langle u, x \rangle^2 + (3C^2 + 4C)\langle u, x \rangle + \ln 2) - 1$$

where $C \approx -2.15$, $u := [1, 1, C - 1]$. Let $f_2(x) := 1$, $f_3(x) := 1$. Then there exists a function $g(z) := \exp(a_3 z^3 + a_2 z^2 + a_1 z + a_0) - 1$, where $a_0, a_1, a_2, a_3 \in \mathbb{R}$ such that for some $w_1, w_2, w_3 \in \mathbb{R}^3$, we have

$$\begin{aligned}
g(\langle w_1, x \rangle) &= f_1(x), \\
g(\langle w_2, x \rangle) &= f_2(x), \\
g(\langle w_3, x \rangle) &= f_3(x),
\end{aligned}$$

Moreover we have

$$g(z) := \exp(z^3 + (-2 - 3C)z^2 + (3C^2 + 4C)z + \ln 2) - 1.$$

and $w_1 = [1, 1, C - 1]^\top$, $w_2 = [0, 0, 0]^\top$, $w_3 = [0, 0, 0]^\top$.

Proof. Let $g(z) := \exp(z^3 + (-2 - 3C)z^2 + (3C^2 + 4C)z + \ln 2) - 1$. Let $w_1 := [1, 1, 1 - C]^\top$, $w_2 := [0, 0, 0]^\top$, $w_3 := [0, 0, 0]^\top$. Then it is clear that we have

$$\begin{aligned}
g(\langle w_1, x \rangle) &= f_1(x), \\
g(\langle w_2, x \rangle) &= f_2(x), \\
g(\langle w_3, x \rangle) &= f_3(x).
\end{aligned}$$

Thus we complete the proof. $\square$

Lemma C.3. *Let*

$$f_1(x) := \exp(\langle u, x \rangle^3 + (-2 - 3C)\langle u, x \rangle^2 + (3C^2 + 4C)\langle u, x \rangle + \ln 2) - 1$$

where $C \approx -2.15$, $u := [1, 1, C - 1]$. It holds that

$$f_1([x_1, 1, 1]^\top) = \exp(x_1^3 - 2x_1^2) - 1.$$

Proof. Note that $\langle u, [x_1, 1, 1]^\top \rangle = x_1 + C$. Hence

$$\begin{aligned}
f_1([x_1, 1, 1]^\top) &= \exp((x_1 + C)^3 + (-2 - 3C)(x_1 + C)^2 + (3C^2 + 4C)(x_1 + C) + \ln 2) - 1 \\
&= \exp(x_1^3 - 2x_1^2) - 1
\end{aligned}$$

where the first step follows from $\langle u, [x_1, 1, 1]^\top \rangle = x_1 + C$, and the second step follows from expanding the higher-order terms and simplifying them. $\square$

C.2 Extend to Neural Networks

We are now prepared to construct a neural network with exponential activation functions. By carefully designing the weight matrix, we can ensure that the network converges to different fixed points for different initialization. We give a 3-d neural network version.

Lemma C.4. *Let $W := [w_1, w_2, w_3] \in \mathbb{R}^{3 \times 3}$ be a weight matrix. Let $C := -2.15$ be a constant. Let a exponential activation function $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as*

$$g(z) := \exp(z^3 + (-2 - 3C)z^2 + (3C^2 + 4C)z + \ln 2) - 1.$$

We define one layer of a neural network, denoted as f , as follows:

$$f(x; W) := [g(\langle w_1, x \rangle), g(\langle w_2, x \rangle), g(\langle w_3, x \rangle)]^\top.$$

Then there exists a weight matrix W following statements hold:

- *The function $f(x; W)$ has at least two fixed points which can be found by the fixed-point iteration, and we denoted them as p_1, p_2 .*
- *For $i \in \{1, 2\}$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in [p_{i,1} - \epsilon_i, p_{i,1} + \epsilon_i] \times \{1\} \times \{1\}$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)}; W)$ converges to the fixed point p_i .*
- *For $i \in \{1, 2\}$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 1)$ such that for any $t \geq 2$,*

$$\|x^{(t)} - p_i\|_\infty \leq K_i^t \cdot c_i \epsilon_i.$$

Proof. The lemma holds by combining Lemma C.1, Lemma C.2, and Lemma C.3. □

C.3 Extension to d -dimensional Case

Next, we extend Lemma C.2 to d -dimensional case.

Lemma C.5. *Let*

$$f_1(x) := \exp(\langle u, x \rangle^3 + (-2 - 3C)\langle u, x \rangle^2 + (3C^2 + 4C)\langle u, x \rangle + \ln 2) - 1$$

where $C \approx -2.15$, $u := [1, \frac{1}{d-1}, \dots, \frac{1}{d-1}, \frac{1}{d-1} + C - 1]^\top \in \mathbb{R}^d$. For all $i \in \{2, \dots, d\}$, let $f_i(x) := 1$. Then there exists a function $g(z) := \exp(a_3 z^3 + a_2 z^2 + a_1 z + a_0) - 1$, where $a_0, a_1, a_2, a_3 \in \mathbb{R}$ such that for some $w_1, w_2, \dots, w_d \in \mathbb{R}^d$, we have that for all $i \in [d]$

$$g(\langle w_i, x \rangle) = f_i(x)$$

Moreover we have

$$g(z) = \exp(z^3 + (-2 - 3C)z^2 + (3C^2 + 4C)z + \ln 2) - 1$$

and $w_1 = [1, \frac{1}{d-1}, \dots, \frac{1}{d-1}, \frac{1}{d-1} + C - 1] \in \mathbb{R}^d$, for all $i \in \{2, \dots, d\}$, $w_i = [0, \dots, 0] \in \mathbb{R}^d$.

Proof. Let $g(z) = \exp(z^3 + (-2 - 3C)z^2 + (3C^2 + 4C)z + \ln 2) - 1$. Let $w_1 := [1, \frac{1}{d-1}, \dots, \frac{1}{d-1}, \frac{1}{d-1} + C - 1] \in \mathbb{R}^d$. For all $i \in \{2, \dots, d\}$, let $w_i = [0, \dots, 0] \in \mathbb{R}^d$. Then it is clear that for all $i \in [d]$ we have

$$g(\langle w_i, x \rangle) = f_i(x)$$

Thus we complete the proof. □